A3Q2

Lily Li

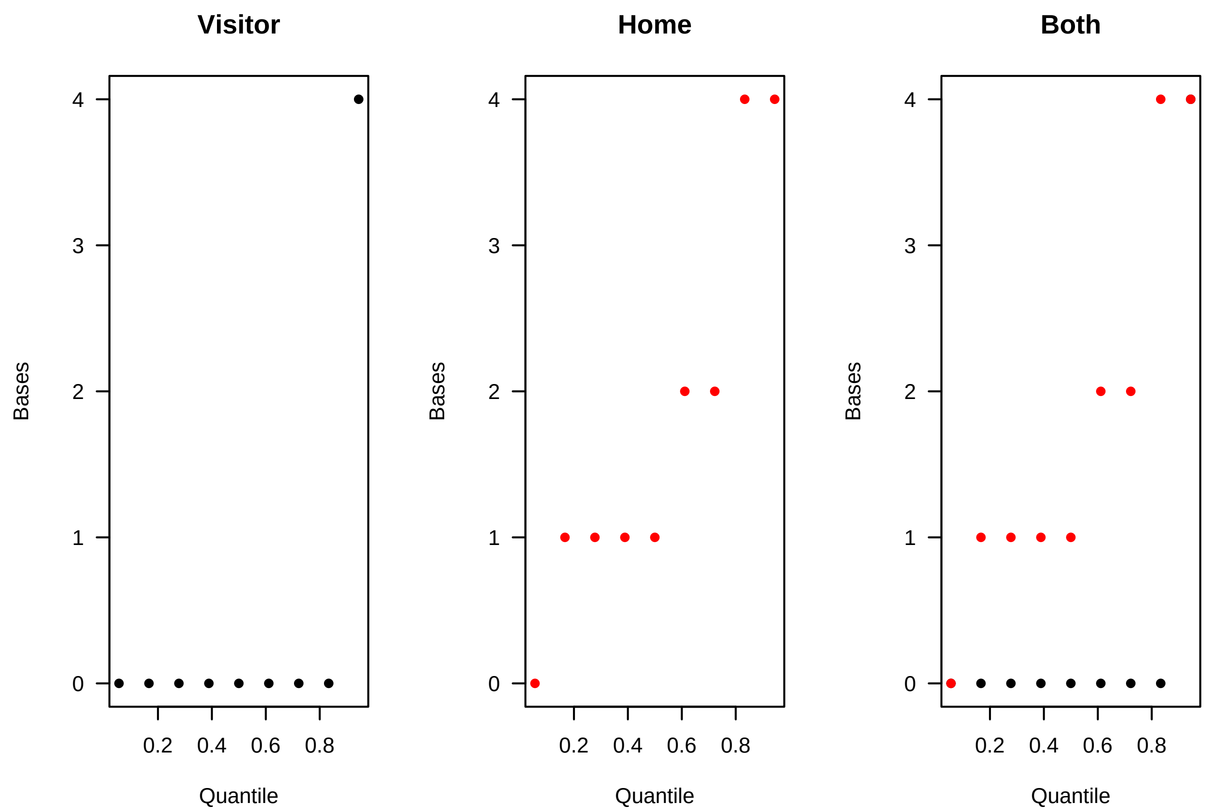
2025-07-05

Q2

(a)

```
mlb_bs <- read.csv("MLB_Batting_Boxscores_2024-04.csv")
mlb_visitor <- mlb_bs[1:9, ]
mlb_home <- mlb_bs[10:18, ]
```

```
yV = sort(mlb_visitor$total_bases)
yH = sort(mlb_home$total_bases)

x = ((1:length(yV)) - 0.5)/length(yV)

par(mfrow = c(1,3))


plot(x, yV, xlab="Quantile", ylab="Bases", pch=16,
las=1, ylim=c(0,max(yV, yH)), main="Visitor")
plot(x, yH, xlab="Quantile", ylab="Bases", pch=16,
las=1, col = "Red", ylim=c(0,max(yV, yH)), main="Home")

plot(x, yV, xlab="Quantile", ylab="Bases", pch=16,
las=1, ylim=c(0,max(yV, yH)), main="Both")
points(x, yH, col="Red", pch=16)
```




```r
par(mfrow = c(1,1))
```

(b)

H0: The total_base distribution for visitors and home batters arefrom the same population.

(c)

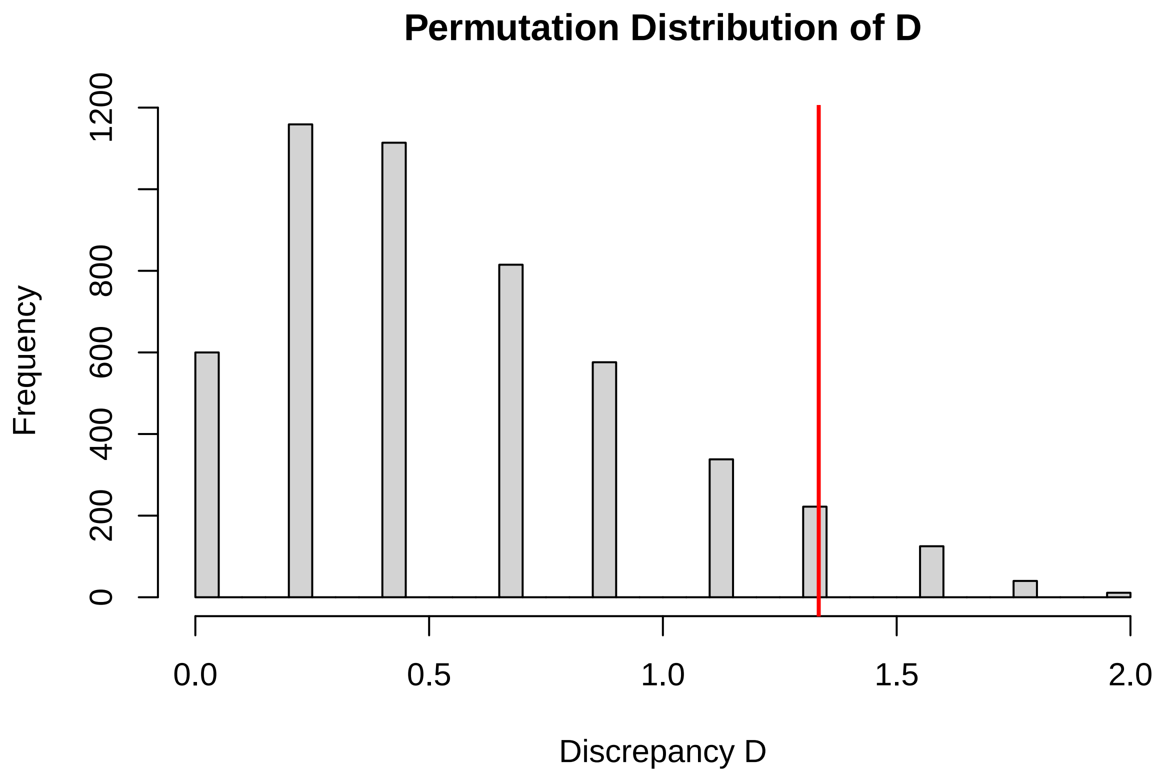
```r
mean_V <- mean(mlb_visitor$total_bases)
mean_H <- mean(mlb_home$total_bases)

D_obs <- abs(mean_V - mean_H)

D_obs
```

```
## [1] 1.333333
```

(d)

```
mixRandomly <- function(pop) {

  pop1 <- pop$pop1
  n_pop1 <- nrow(pop1)
  pop2 <- pop$pop2
  n_pop2 <- nrow(pop2)
  mix <- rbind(pop1,pop2)
  select4pop1 <- sample(1:(n_pop1 + n_pop2),
  n_pop1,
  replace = FALSE)
  new_pop1 <- mix[select4pop1,]
  new_pop2 <- mix[-select4pop1,]
  list(pop1=new_pop1, pop2=new_pop2)

}
```

```
pop_list <- list(pop1 = mlb_visitor, pop2 = mlb_home)

M <- 5000
D_star <- numeric(M)

set.seed(341)
for(i in 1:M) {
  mixed <- mixRandomly(pop_list)
  yV_star <- mixed$pop1$total_bases
  yH_star <- mixed$pop2$total_bases
  D_star[i] <- abs(mean(yV_star) - mean(yH_star))
}

hist(D_star, breaks=30, main="Permutation Distribution of D", xlab="Discrepancy D")
abline(v=D_obs, col="red", lwd=2)
```



(e)

```
p_value_approx <- mean(D_star >= D_obs)
p_value_approx
```

```
## [1] 0.0796
```

(f)

Since the p-value exceeds 0.05, we do not have sufficient evidence to reject H0. We conclude that there is no significant difference in total bases between the visitor and home player populations.

(g)

```
combos <- combn(18, 9)
PV_10000 <- combos[,10000]
PH_10000 <- setdiff(1:18, PV_10000)

PV_10000
```

```
## [1]  1  2  6  7  9 11 12 15 16
```

```
PH_10000
```

```
## [1]  3  4  5  8 10 13 14 17 18
```

(h)

```
D_all <- numeric(ncol(combos))

y_all <- mlb_bs$total_bases[1:18]

for (i in 1:ncol(combos)) {
  PV_idx <- combos[,i]
  PH_idx <- setdiff(1:18, PV_idx)
  D_all[i] <- abs(mean(y_all[PV_idx]) - mean(y_all[PH_idx]))
}
  p_value_exact <- mean(D_all >= D_obs)
  p_value_exact
```

```
## [1] 0.08605512
```

(i)

There is not substantial difference between the results from part (h) and here, this is not surprising because 5000 permutations is large enough to approximate the exact distribution closely.